

Insulin Inhibits Dog Vascular Smooth Muscle Contraction and Lowers Ca^{2+}_i by Inhibiting Ca^{2+} Influx¹

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ABSTRACT Essential hypertension, obesity and non-insulin-dependent diabetes are associated with resistance to insulin-induced glucose disposal. Because physiological concentrations of insulin inhibit vascular smooth muscle (VSM) contraction in vivo, it has been proposed that resistance to insulin-induced inhibition of VSM contraction might be partly responsible for the elevated vascular resistance found in these clinical conditions. Nevertheless, it is not known how insulin inhibits contraction of normal VSM. Several workers have demonstrated that insulin attenuates the agonist-induced intracellular Ca^{2+} (Ca^{2+}_i) transient in VSM, although the identity of the sarcolemmal and/or sarcoplasmic reticular Ca^{2+} transport systems that are affected by insulin is controversial. Our laboratory has examined the effects of a physiological concentration of insulin on contraction and Ca^{2+} transport in cultured VSM cells from canine femoral artery. We have found that insulin inhibits agonist-induced contraction and attenuates the agonist-induced Ca^{2+}_i transient by inhibiting Ca^{2+} influx, but not by increasing Ca^{2+} efflux or inhibiting Ca^{2+} release from internal stores. Insulin also stimulates ouabain-sensitive $^{86}\text{Rb}^+$ uptake ($\text{Na}^+\text{-K}^+$ pump activity) and does not inhibit VSM contraction in the presence of ouabain. Our data support the hypothesis that insulin stimulates $\text{Na}^+\text{-K}^+$ pump activity, resulting in hyperpolarization of the cell and decreased Ca^{2+} influx via voltage-operated channels. *J. Nutr.* 125: 1732S-1737S, 1995.

INDEXING KEY WORDS:

- hypertension • diabetes • obesity • serotonin
- $\text{Na}^+\text{-K}^+$ pump

Increased blood pressure is associated with human obesity and noninsulin-dependent diabetes. These conditions, as well as essential hypertension, are associated with resistance to insulin-stimulated total body glucose disposal (Epstein and Sowers 1992, Ferrannini et al. 1987). Resultant hyperinsulinemia occurs as the pancreas attempts to maintain plasma glucose concentration in the normal range. Investigators have proposed that the resultant hyperinsulinemia could

contribute to the hypertensive state by increasing renal Na^+ reabsorption and sympathetic nerve activity (Epstein and Sowers 1992). However, neither chronic infusion of insulin in dogs (Hall et al. 1990) nor insulinoma in man (Tsutsu et al. 1990) is associated with hypertension. Thus, the reasons for the relationship between hyperinsulinemia and hypertension are still controversial.

Insulin resistance hypothesis

It is noteworthy that most studies that have examined the acute hemodynamic effects of insulin in man have demonstrated that physiological concentrations of insulin cause vasodilation (Anderson et al. 1991, Doria et al. 1991, Laakso et al. 1990, Laakso et al. 1992). In patients with noninsulin-dependent diabetes mellitus (Laakso et al. 1992), in obese subjects (Laakso et al. 1990) and in patients with essential hypertension (Doria et al. 1991), resistance to insulin's glucose-lowering effect is accompanied by resistance to insulin's vasodilatory effect. These data suggest the following hypothesis: insulin normally inhibits vascular smooth muscle contraction, and in conditions where peripheral tissues are insensitive to insulin's glucose-lowering effect, vascular smooth muscle is also resistant to insulin's attenuation of tonic neural and humoral mediator-induced mechanisms that increase vessel tone. This insulin resistance would then result in abnormally high vessel tone, which could contribute to the hypertensive state. This hypothesis is supported by a recent study, which reported that resistance to insulin-induced vasodilation of a dorsal hand vein is present in hypertensive patients. Body mass index in-

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dependently correlated with resistance to insulin-induced venodilation (Feldman and Bierbrier 1993). This hypothesis is also supported by animal data. Insulin-resistant obese Zucker rats exhibited hypertension associated with exaggerated reactivity of aortic strips to contractile agonists (Zemel et al. 1991). Even though insulin attenuated phenylephrine-induced contraction in both the lean and obese aortas, aortas from lean rats exhibited about a threefold greater attenuation in contraction than aortas from obese rats (Zemel et al. 1991). These data suggest that increased vascular reactivity in hypertensive obese Zucker rats may result from their insulin-resistant state and, consequently, a diminished ability of insulin to attenuate agonist-induced vasoconstriction.

Mechanism of insulin effect

The sites and mechanisms by which insulin attenuates vasoconstriction are unknown. Although a direct effect of insulin on VSM is possible, it has been reported that beta-adrenergic blockers inhibit insulin's vasodilatory effect in man and dog, suggesting that the sympathetic nervous system is involved (Creager et al. 1985, Liang et al. 1982).

It has recently been reported that insulin (100 mU/ml) inhibited a vasopressin-induced and depolarization-induced inward current (presumably Ca^{2+} current) in a7r5 cells and attenuated the peak height of the vasopressin-induced Ca^{2+}_i transient (Standley et al. 1991). Insulin (100 mU/ml) increased the rate of decline of Ca^{2+}_i after its rise after stimulation with vasopressin in a7r5 cells (Standley et al. 1991). In addition, insulin (100 $\mu\text{U}/\text{ml}$) increased $^{45}\text{Ca}^{2+}$ efflux from unstimulated rat aortic strips and increased their rate of relaxation after phenylephrine-induced contraction (Zemel et al. 1992). Other workers have reported that insulin (100 mU/ml) attenuated angiotensin II (AII)-induced Ca^{2+} mobilization from IP_3 -sensitive intracellular stores in cultured rat VSM cells (Saito et al. 1993). Individually, these experiments support the various possibilities that in rat vascular smooth muscle (VSM), insulin inhibits agonist-stimulated VSM contraction by inhibiting Ca^{2+} influx via Ca^{2+} channels, by stimulating sarcolemmal (SL) Ca^{2+} efflux or by inhibiting Ca^{2+} mobilization from intracellular stores. The physiological relevance of these findings is uncertain because cultured VSM cells have usually changed from a contractile to a synthetic phenotype (Campbell et al. 1988), and the insulin concentrations used in most of these experiments are much higher than physiological concentrations (<100 $\mu\text{U}/\text{ml}$) (Saito et al. 1993, Standley et al. 1991). These concentrations of insulin could activate other receptors such as the insulin-like growth factor 1 receptor (Flier et al. 1986).

It has recently been reported that acute (1-hour) exposure to insulin (100 mU/ml) increased the mRNA

for the alpha-2 subunit of the $\text{Na}^+\text{-K}^+$ pump in a7r5 cells and stimulated $\text{Na}^+\text{-K}^+$ ATPase (enzymatic) activity in membranes isolated from those cells (Tirupattur et al. 1993). Others have reported that physiological concentrations of insulin stimulate $\text{Na}^+\text{-K}^+$ pump activity in skeletal muscle (Hundal et al. 1992, Rosic 1985). In this regard it is noteworthy that $\text{Na}^+\text{-K}^+$ pump activity in vascular smooth muscle modifies contraction. Stimulating the pump hyperpolarizes the cell, thereby inhibiting Ca^{2+} influx via voltage-operated channels (VOCs). Studies have shown that inhibiting the $\text{Na}^+\text{-K}^+$ pump in VSM with ouabain or zero extracellular K^+ (K^+_o) depolarized and contracted the tissue. Returning normal K^+ concentration to the bathing media of zero K^+_o -contracted vessels reactivated the $\text{Na}^+\text{-K}^+$ pump, hyperpolarized the cells and promptly relaxed the tissue (Anderson 1976, Anderson et al. 1983). It seems reasonable that insulin's inhibition of vascular smooth muscle contraction could be related to stimulation of $\text{Na}^+\text{-K}^+$ pump activity by insulin.

We sought to examine the relationship between insulin, contraction and ion transport in individual cells using canine femoral artery vascular smooth muscle cells cultured in a low concentration (0.5%) of fetal calf serum (FCS). Cells do not proliferate when grown under these conditions but maintain the ability to contract (Kahn et al. 1992). We found that a physiological concentration of bovine insulin (40 $\mu\text{U}/\text{ml}$) acutely (20 minutes preincubation) inhibited serotonin (5-HT) and AII-stimulated contraction of individual cells (**Fig. 1**) (Kahn et al. 1993). These results were consistent with previous studies, which showed that insulin acutely causes vasodilation in man (Anderson et al. 1991, Doria et al. 1991, Laakso et al. 1990) and dog (Hall et al. 1990) and inhibits contraction of rat arterial tissue in vitro (Zemel et al. 1992). Vasodilation could be explained by insulin's inhibition of tonic neural and humoral mediator-induced mechanisms that are responsible for maintaining vessel tone. However, proposing that insulin-resistant vessels are chronically overcontracted compared with normal vessels requires demonstrating that chronic exposure to a physiological insulin concentration inhibits agonist-induced contraction in normal vessels. Cells were preincubated with and without 40 $\mu\text{U}/\text{ml}$ insulin for 1 week. We found that this also inhibited 5-HT or AII-induced contraction (Kahn et al. 1993).

Individual VSM cells were plated on collagen gels and loaded with the Ca^{2+} -sensitive fluorescent indicator FURA II. We found that 40 $\mu\text{U}/\text{ml}$ insulin (20-minute preincubation) attenuated the 5-HT-induced Ca^{2+}_i transient and inhibited contraction in the same individual VSM cells (**Fig. 2**) (Kahn et al. 1993).

To determine if contraction because of depolarization-induced Ca^{2+} influx was attenuated by insulin, cells were preincubated for 20 minutes with

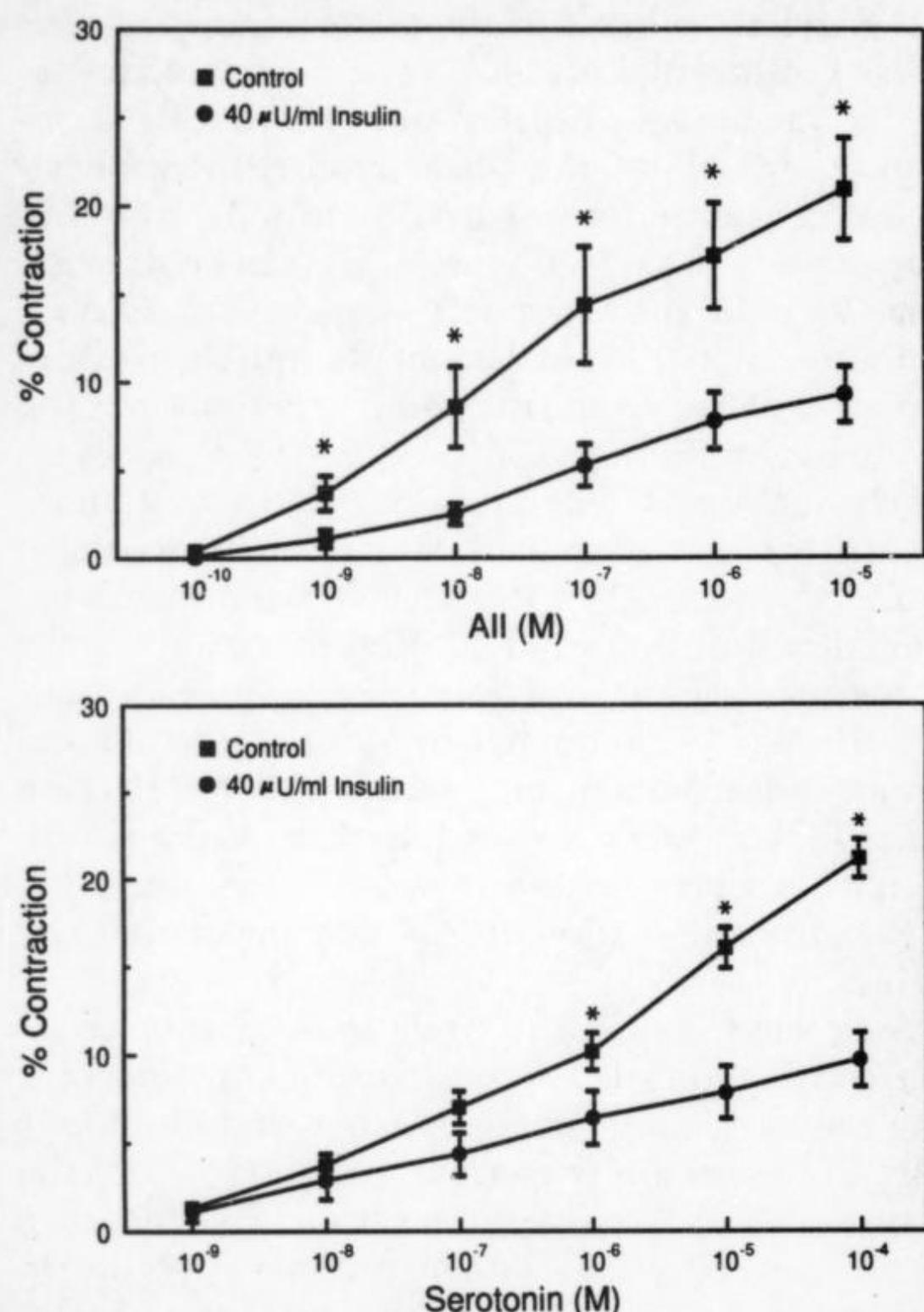


FIGURE 1 Effect of acute insulin exposure on agonist-stimulated contraction of vascular smooth muscle. Dishes of cells were preincubated for 20 minutes at 37°C with or without 40 μ U/ml insulin in HEPES PSS plus 4% Bovine serum albumin (BSA) and a baseline photomicrograph obtained. The lowest indicated concentrations of All (top) or 5-HT (bottom) were added to the bathing media with or without insulin and another photomicrograph obtained at 10 minutes. Sequentially higher concentrations of agonists were added to the bathing media and subsequent photomicrographs obtained 10 minutes after each increment of agonist. Data points represent the mean $\pm$ SE of the mean contractions from the baseline lengths of 6–10 cells in each of six dishes (* $P < 0.05$).

or without insulin (40 μ U/ml) and then acutely exposed to high K_o^+ . High K_o^+ caused contraction in a concentration-dependent manner, presumably because of depolarization-stimulated Ca^{2+} influx via voltage-operated channels (VOCs). The maximal effect of high extracellular K^+ -induced contraction was not affected by acute insulin exposure, but the ED_{50} for K_o^+ -induced contraction was increased by insulin from 7.6 ± 2.5 to 16.0 ± 3.9 mM ($P < 0.05$) (Fig. 3) (Kahn et al. 1993). These data are consistent with the possibility that insulin directly increases the voltage threshold (to more depolarized values) for Ca^{2+} influx via VOCs. Alternatively, they are also consistent with the notion that insulin primarily hyperpolarizes the cell (by stimulating Na^+ - K^+ pump

activity for instance). As a result, depolarization of the cell by moderate increases in K_o^+ would be offset by insulin. This would shift the ED_{50} for K_o^+ -induced contraction to a higher value.

If insulin's attenuation of agonist-induced contraction is caused by inhibition of Ca^{2+} influx through VOCs, insulin should have no effect in the presence of VOC blockade. This was found to be the case. Verapamil (10^{-6} M), added at the same time as 5-HT (10^{-5} M), partially inhibited 5-HT-induced contraction, but insulin did not inhibit contraction further (Kahn et al. 1993). Similar results were obtained when 10^{-6} M All was used instead of 5-HT. Because agonist-stimulated contraction in the presence of verapamil depends on Ca^{2+} release from intracellular stores after stimulation of the signal transduction system cascade, these data also indicate that insulin does not affect the individual components of the contractile agonist-stimulated signal transduction system or Ca^{2+} release from intracellular stores in our VSM cell model.

The data from our laboratory presented thus far are consistent with the possibilities that insulin directly inhibits VOC activity and/or that insulin hyperpolarizes the cell, thereby secondarily inhibiting VOC activity. In that regard, it has been reported that several vasodilators inhibit VSM contraction by hyperpolarizing the cell via increasing K^+ channel activity (Brayden et al. 1991). To test whether insulin increases K^+ channel activity in our VSM cells, dishes of cells were preloaded with $^{86}Rb^+$ for 3 hours with or without 40 μ U/ml insulin for the last 20 minutes. The loading solution was replaced with $^{86}Rb^+$ -free solution in the presence and absence of 10^{-5} M 5-HT, and the rate of $^{86}Rb^+$ efflux was assayed. We found that 5-HT increased the first order rate constant for $^{86}Rb^+$ efflux from 0.054 ± 0.003 to 0.074 ± 0.003 min $^{-1}$ ($P < 0.01$). This finding was consistent with other workers' findings that contractile agonists increase the activity of K^+ channels in VSM. We also found that insulin had no effect on either of these values (Kahn, A., unpublished results). We concluded from this experiment that insulin does not increase Rb^+ permeability (a measure of K^+ permeability) and therefore could not hyperpolarize our cells by opening K^+ channels.

As mentioned above, stimulation of the electrogenic Na^+ - K^+ pump in VSM can also result in cell hyperpolarization (Anderson 1976), and physiological concentrations of insulin stimulate the activity of the Na^+ - K^+ pump and hyperpolarize skeletal muscle cells (Hundal et al. 1992, Rosic et al. 1995, Zierler 1988). To see if insulin stimulated the activity in the VSM Na^+ - K^+ pump, ouabain-sensitive $^{86}Rb^+$ uptake was measured in our nonproliferated canine femoral artery VSM cells (Kahn et al. 1993). As shown in Fig. 4, preincubating cells for 20 minutes with 40 μ U/ml insulin increased the 10-minute ouabain-sensitive uptake to 247% of control ($P < 0.05$); 10^{-5} M 5-HT alone stimulated it to 248% of control ($P < 0.05$). Insulin

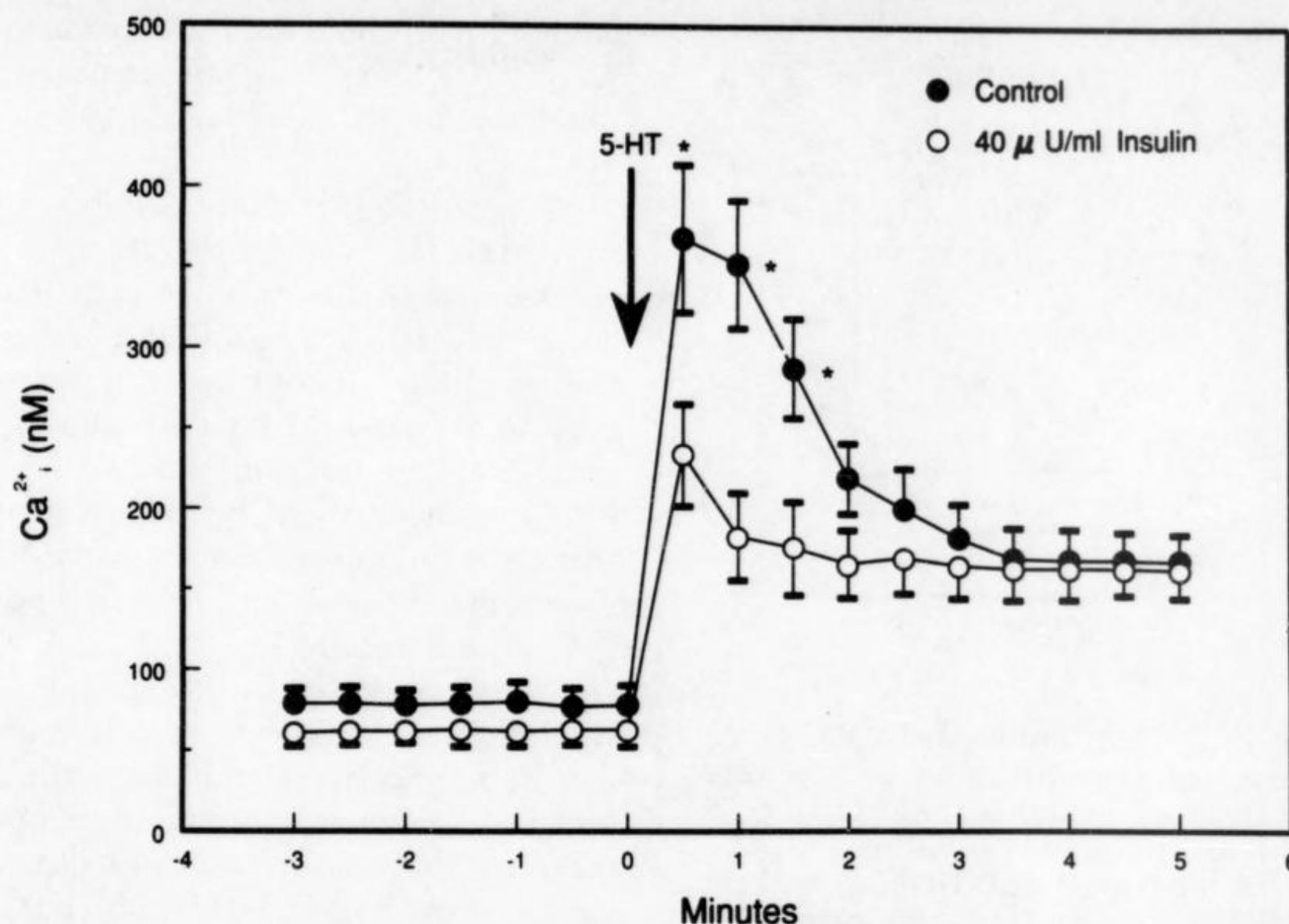


FIGURE 2 Effect of insulin and 5-HT on Ca_i^{2+} in individual vascular smooth muscle cells. Cells were loaded with FURA II, superfused with HEPES PSS with or without 40 $\mu\text{U/ml}$ insulin for 20 minutes. 5-HT (10^{-5} M) was added to the superfusion solution, and the fluorescence emission from a single cell was continuously monitored. Data points represent the mean $\pm$ SE of Ca_i^{2+} values from seven cells each with or without insulin (* $P < 0.05$).

plus 5-HT stimulated ouabain-sensitive $^{86}\text{Rb}^+$ uptake to 373% of control ($P < 0.05$ vs. insulin alone or 5-HT alone). These data demonstrate that 5-HT alone stimulates Na^+/K^+ pump activity in nonproliferated canine femoral artery VSM cells. The same response

has been described in confluent primary cultures of these cells (Navran et al. 1990). More importantly, these data show that a physiological insulin concentration independently stimulates Na^+/K^+ pump activity in these VSM cells. We propose that this may ultimately be responsible for insulin's attenuation of agonist-induced contraction.

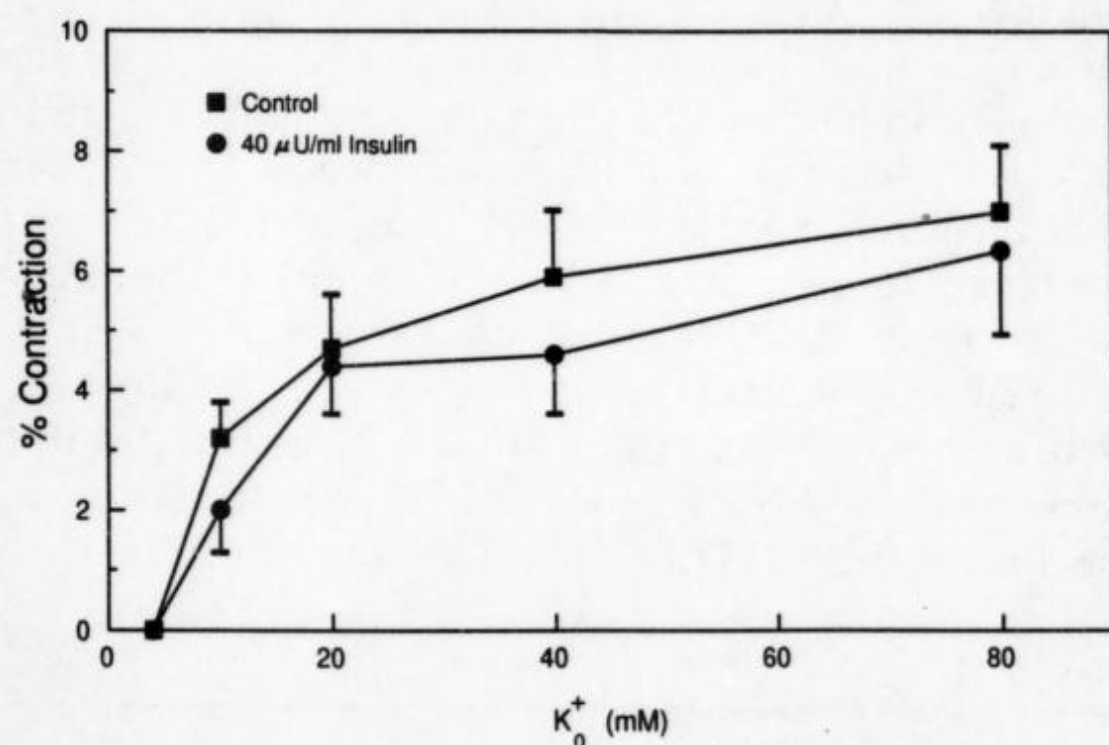


FIGURE 3 Effect of insulin on K^+ -induced contraction of individual vascular smooth muscle cells. Dishes of cells were preincubated for 20 minutes with or without 40 $\mu\text{U/ml}$ insulin in HEPES PSS plus 4% BSA and a baseline image obtained. The bathing media were removed and replaced with HEPES PSS plus 4% BSA, which contained the indicated concentrations of K^+ (isototically substituted for Na^+) with or without 40 $\mu\text{U/ml}$ insulin. Another photomicrograph was taken after 10 minutes. Data points represent the mean $\pm$ SE of the mean contractions from baseline lengths of 6–10 cells in each of five dishes for each K^+ concentration.

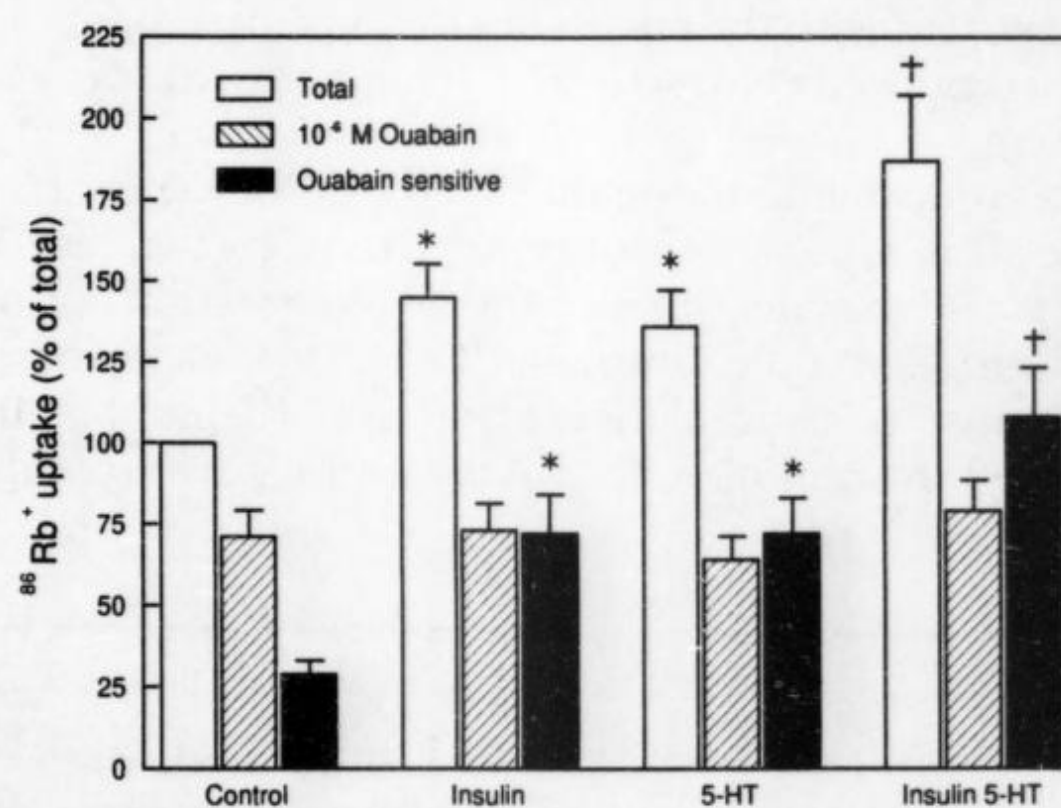


FIGURE 4 Effect of insulin and 5-HT on $^{86}\text{Rb}^+$ uptake in VSM cells. Cells were grown in Dulbecco's modified Eagle's medium (DMEM) plus 0.5% FCS for 4–7 days. They were preincubated with or without insulin (40 $\mu\text{U/ml}$) for 20 minutes in HEPES PSS, and the 10-minute uptake of $^{86}\text{Rb}^+$ was assayed in the presence and absence of ouabain (1 μM) and in the presence and absence of 5-HT (10^{-5} M). * $P < 0.05$ vs. control. + $P < 0.05$ vs. 5-HT or insulin alone.

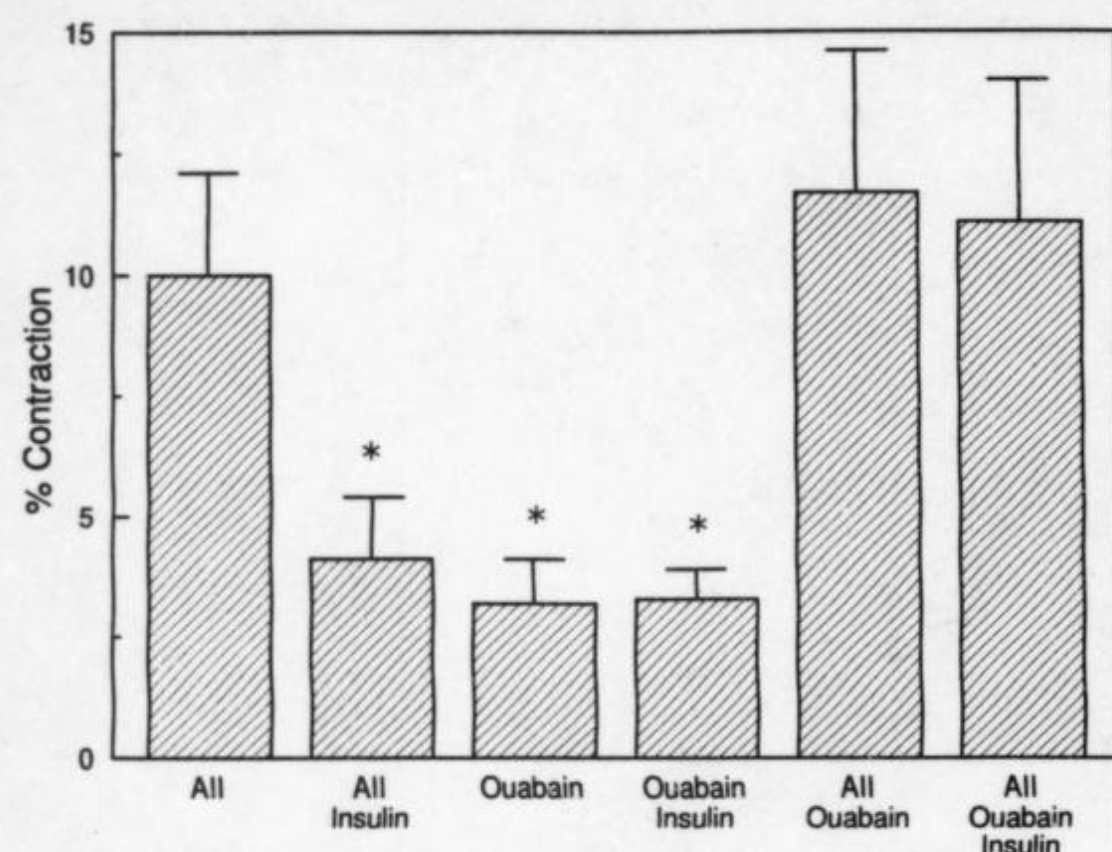


FIGURE 5 Effect of ouabain, insulin and All on contraction of individual vascular smooth muscle cells. Dishes of cells were preincubated for 20 minutes at 37°C in HEPES PSS plus 4% BSA with or without 40 μ U/ml insulin and a baseline photomicrograph obtained. All (10^{-6} M), ouabain (10^{-6} M) or All (10^{-6} M) plus ouabain (10^{-6} M) were added to the bathing media and another photomicrograph obtained 10 minutes later. Data represent the mean $\pm$ SE of the mean contractions of 6–10 cells in each of five dishes (* $P < 0.05$ vs. All alone).

To test this, we measured the effect of insulin on agonist-induced VSM cell contraction in the presence and absence of ouabain, an inhibitor of the $\text{Na}^+\text{-K}^+$ pump. As shown in **Figure 5**, insulin did not inhibit 10^{-6} M All-induced contraction in the presence of ouabain. Similar results were obtained when 10^{-5} M 5-HT was used instead of All. Ouabain, by itself, induced a small contraction, and insulin did not block this effect either (Kahn et al. 1993). These data are consistent with the possibility that insulin inhibits agonist-induced contraction by stimulating the $\text{Na}^+\text{-K}^+$ pump.

Our data with individual VSM cells of the contractile phenotype are consistent with the view that insulin attenuates agonist-induced VSM contraction by inhibiting Ca^{2+} influx through VOCs. They also indicate that insulin's stimulation of the $\text{Na}^+\text{-K}^+$ pump is involved in insulin's effect. A hypothesis we have de-

veloped with these data is that insulin increases $\text{Na}^+\text{-K}^+$ pump activity in agonist-stimulated cells, resulting in relative hyperpolarization and decreased Ca^{2+} influx via VOCs.

We also recently examined how insulin affects the individual Ca^{2+} transport systems in primary cultured canine femoral artery VSM cells that were grown to confluence in 10% FCS (Kahn et al. 1994). Although these cells do not contract and obviously could not be used in definitive studies to examine how insulin inhibits VSM contraction, their use has provided insight into the mechanism by which insulin affects Ca^{2+} transport in our nonproliferated VSM model that maintains the contractile phenotype.

Ca_i^{2+} was monitored via FURA II fluorescence in confluent cells that had been preincubated with or without 40 μ U/ml insulin for 20 minutes. As was true for the nonproliferated (contractile phenotype) cells, 10^{-5} M 5-HT in Ca^{2+} replete media induced a Ca^{2+} transient that was attenuated by insulin. However, in Ca^{2+} -free media, the 5-HT-induced Ca_i^{2+} transient was reduced and insulin did not reduce it further (Kahn et al. 1994). These data indicated that under control conditions, the 5-HT-induced Ca^{2+} transient is caused by stimulation of Ca^{2+} influx and Ca^{2+} release from intracellular stores. The data also suggested that insulin attenuates the 5-HT-induced Ca_i^{2+} transient by inhibiting Ca^{2+} influx but not by inhibiting Ca^{2+} release from intracellular stores or stimulating sarcolemmal Ca^{2+} efflux.

We verified that insulin inhibited 5-HT-induced Ca^{2+} influx by demonstrating that insulin inhibited 5-HT-induced Mn^{2+} influx [a Ca^{2+} surrogate whose intracellular concentration was monitored by its quenching of FURA II fluorescence at the Ca^{2+} isosbestic wave length] (Kahn et al. 1994). We verified that insulin had no effects on sarcolemmal Ca^{2+} efflux or Ca^{2+} release from intracellular stores by showing that insulin did not affect 5-HT-stimulated $^{45}\text{Ca}^{2+}$ efflux from intact cells and from saponin-skinned cells, respectively (Kahn et al. 1994) (**Table 1**).

In summary, data from our laboratory demonstrate that a physiological concentration of insulin inhibits 5-HT, All and high K_o^+ -induced VSM contraction and attenuates the 5-HT-induced Ca_i^{2+} transient. The latter

TABLE 1
Effect of 5-HT and insulin on $^{45}\text{Ca}^{2+}$ efflux from vascular smooth muscle cells

	$^{45}\text{Ca}^{2+}$ efflux first order rate constant (min^{-1})			
	Control	Insulin	5-HT	5-HT + insulin
No Saponin (n = 6)	0.11 $\pm$ 0.01	0.13 $\pm$ 0.02*	0.25 $\pm$ 0.01†	0.24 $\pm$ 0.02*
Saponin (n = 6)	0.18 $\pm$ 0.01	0.18 $\pm$ 0.02*	0.27 $\pm$ 0.02†	0.25 $\pm$ 0.03*

* $P = \text{NS}$ vs. the same conditions without insulin.

† $P < 0.05$ vs. the same conditions without 5-HT.

is caused by insulin's inhibition of Ca^{2+} influx. Finally, insulin stimulates Na^+ - K^+ pump activity, which may be indirectly responsible for the inhibition by insulin of agonist-induced contraction.

Future studies should reveal whether VSM cells from genetic hypertensive animals or humans are resistant to insulin's inhibition of Ca^{2+} influx and contraction. If such is the case, it would strongly support the hypothesis that resistance to insulin-induced vasodilation is responsible, at least in part, for hypertension in patients with resistance to insulin-induced glucose disposal.

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